Microbiology LIMS POC — Process Reference (with Block Diagrams)

v3.2 (Conditioned Discovery) | Visual companion to `microbio_lims_architecture.drawio` | 2026-04-29

What the system does

A subject-matter expert (SME) drops microbiology SOPs into S3. An agent pipeline, grounded in a hybrid retrieval substrate (BM25 + dense), reads them and emits two artifacts: **(A)** a Jira-ready Epic / Story tree, and **(B)** per-culture YAML configuration profiles (urine, blood, target-pathogen).

The Cytology Jira backlog is used as a *teaching corpus* (few-shot exemplars), not a registry to match against. The agent stops at Story — Tasks (test IDs, library versions, DB columns) are dev-authored once the team has codebase visibility.

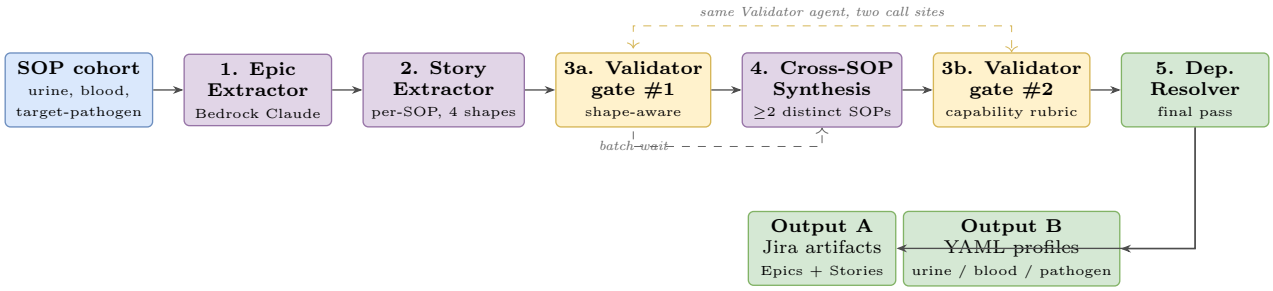
v3.2 addition — Conditioned Discovery. v3.1 generated epics from-scratch and treated themes G1–G8 as a closed Cyto-derived vocabulary. v3.2 adds a *warm-start* step at both levels: a new pre-flight **Theme Discovery agent** bootstraps a discipline's theme catalog by classifying its SOPs against the prior taxonomy first and surfacing only the residual as novel; the **Epic Extractor runs in conditioned mode**, classifying drafts against the prior epic catalog (`cyto_epic_v1`) before clustering anything new. The result: most themes and epics inherit by construction with explicit lineage, and only genuinely-novel structure (with cluster evidence) goes to SME for ratification. See the v3.2 sections below.

Color legend: `src` source/data `cmp` compute `ai` AI/LLM agent `val` validator/decision `store` storage/output `xc` cross-cutting

Key terms (glossary)

SOP	Standard Operating Procedure — the source document the agent reads.
Epic	A capability domain; maps to a Jira Epic. E.g. *Microbiology Specimen Receipt*.
Story	A dev-actionable unit emitted by the agent. Each Story declares one of four `shape` values. Maps to a Jira Story.
Shape	Categorization assigned to each Story at extraction time. Drives Validator routing and downstream interpretation. Four values: `capability`, `workflow-stage-split`, `configuration-instance`, `cleanup`.
Theme (G1–G8)	Eight content tags spanning the LIMS workflow: G1 Pre-Analytic, G2 Analytic, G3 Post-Analytic, G4 Reporting, G5 QC, G6 Compliance, G7 Instrumentation, G8 Platform. Multi-label per chunk.
Teaching corpus	Curated SME-validated (SOP excerpt → Story) pairs from Cyto. Used as few-shot exemplars at extraction time.
Retrieval slot	A parallel retrieval channel scoped to a particular discipline + theme combination. Up to 5 slots fire per query.
Intent	Structured interpretation of a user query: {`action, themes, discipline, granularity, refs, confidence`}.
τ **(tau)**	Intent-confidence threshold (0.7). Below this, the system asks a clarifying question rather than retrieving.
θ **(theta)**	Retrieval-relevance threshold (0.5). If every slot falls below this, the system refuses to generate.
Batch wait	The pipeline pause between Validator gate #1 and Cross-SOP Synthesis. Synthesis cannot run until every per-SOP Story has cleared gate #1.
v3.2 additions:	
Conditioned discovery	A four-pass procedure (classify → cluster residual → SME ratify → re-tag) that warm-starts a new catalog against an existing prior, rather than clustering from scratch. Applied at both theme and epic levels.
Prior catalog	The existing themes (or epics) that conditioned discovery treats as the structure-to-reuse. Cyto's catalogs are the prior for both Micro and Histology bootstraps.
Residual	Inputs whose classification confidence falls below τ_{match} in Pass 1. Only residuals are clustered for novelty.
τ_{match}	Match-confidence threshold for Pass 1 (default 0.65). Distinct from the query-time τ above. Higher $\tau_{match} \Rightarrow$ eager-to-inherit.
$\varepsilon_{novelty}$	Minimum cluster size in Pass 2 (default 3). Higher ⇒ stricter evidence bar before admitting a novel theme/epic.
G0 / E0	First-class *unclassified* bucket on every theme catalog (G0) and epic catalog (E0). 5% alarm threshold over a rolling window triggers re-running discovery.
Catalog versioning	Theme catalogs are named `<discipline>_v<N>` (e.g. `cyto_v1`, `histo_v1`); epic catalogs are named `<discipline>_epic_v<N>` (e.g. `cyto_epic_v1`). They version independently. Stories carry a `theme_catalog_version` field for theme resolution.
ANALOGY map	Explicit cross-catalog link table. Each link typed: `identical` / `partial` / `discarded_in_target` / `novel_in_target`. Replaces v3.1's implicit shared-theme assumption.

Agent pipeline — 5 agents, Validator gates twice



1. **Epic Extractor** — identifies capability domains. *Out:* Epic record per domain.
2. **Story Extractor** — per-SOP, multi-shape. Different elements of one SOP can take different shapes.
3. **Validator gate #1** — shape verification + shape-specific sub-rubric. Two revisions then SME-escalate.
4. **Cross-SOP Synthesis** — runs after gate #1 clears the entire batch. Behavioral-similarity clustering over (title + AC + source excerpt); cluster qualifies only if its members come from ≥ 2 *distinct* SOPs. Lifts a capability Story per cluster, with `cross_links[]` back to the contributing concretes.
5. **Validator gate #2** — same Validator agent as gate #1, second call site; capability rubric only. Synthesized stories are a *new* artifact, not a derivative, so they re-pass the bar.
6. **Dependency Resolver** — story↔story dependencies, stage-split sibling links, capability↔child links. Finalizes the artifact graph for Jira push.

v3.2 modifications inside the main pipeline: the Epic Extractor (1) gains *conditioned mode* — it classifies its drafts against the prior epic catalog before clustering anything as novel. The other four agents are unchanged in shape. A *new sixth agent* — the **Theme Discovery agent** — runs as pre-flight (one-time per discipline), not on the per-SOP critical path. Both are detailed in the next two sections.

v3.2 Theme Discovery — pre-flight, one-time per discipline

Theme Discovery runs *before* the main pipeline boots for a new discipline (and again only when the G0 alarm trips). Its job: emit a versioned theme catalog (e.g. `histo_v1`) by warm-starting against an existing prior (e.g. `cyto_v1`). Most themes inherit; novel ones must clear an evidence bar and an SME gate.

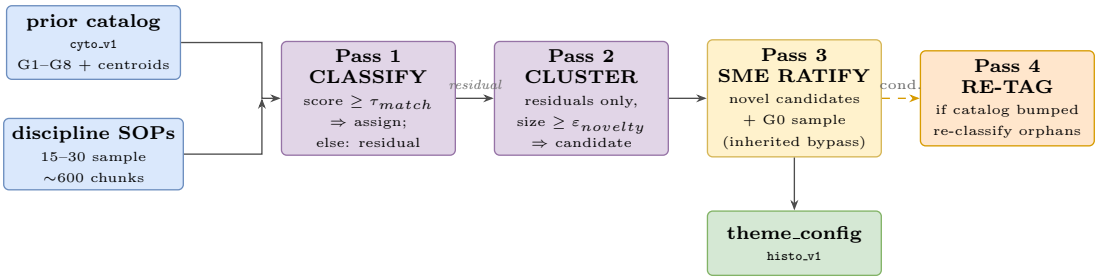


Figure 1: Theme Discovery — four passes, pre-flight per discipline. Pass 4 fires only when SME ratification changes the catalog (novel ratified or theme discarded).

Algorithm in pseudocode:

```
PASS 1 (CLASSIFY): for each chunk:  score = best_match(chunk, prior); if score ≥ τ_match assign else residual
PASS 2 (CLUSTER): cluster residuals (HDBSCAN); each cluster of size ≥ ε_novelty → candidate_novel_theme
PASS 3 (SME GATE): SME confirms / renames / rejects each novel candidate; inherited themes pass automatically
PASS 4 (RE-TAG, if catalog changed):  re-classify previously-tagged corpus chunks against catalog_v(N+1)
```

Two knobs control the bias. τ_{match} (default 0.65): higher ⇒ eager-to-inherit (small residual; risk forced fits). $\varepsilon_{novelty}$ (default 3): higher ⇒ stricter evidence bar before admitting novelty. The user's stated motive — "don't just end up with entirely new ones" — corresponds to a high τ_{match} and a non-trivial $\varepsilon_{novelty}$.

Worked example — bootstrapping Histology from `cyto_v1`. Pass 1 classifies 460/600 chunks into G1–G8 (residual = 140, including 12 chunks that fit G7 above τ_{match}). Pass 2 finds 4 clusters above $\varepsilon_{novelty}$: H1 Tissue Processing (38), H2 Staining (32), H3 Block Archive (25), H4 Frozen Section (23); 22 chunks → G0 (3.7%, below alarm). Pass 3: SME confirms all 4 novel themes; flags G7 as redundant in histo (instrument concerns fold into H1) — discarded. Pass 4 re-classifies the 12 G7-orphaned chunks against the post-discard taxonomy (most land in H1, consistent with the ANALOGY map's `G7→H1, partial` link). Final `histo_v1`: 11 active themes (7 inherited + 4 novel; G7 recorded in ANALOGY map only). Reuse rate: 7/11 ≈ 64%; 7/8 of the prior themes (87.5%) survived.

v3.2 Conditioned Epic Extractor — match before generate

In v3.1 the Epic Extractor proposed Micro epics from-scratch and optionally annotated `cyto_epic_analog` post-hoc when retrieval signal happened to align. In v3.2 the order is reversed: drafts are classified against the prior epic catalog *first*; matched drafts inherit the analog by construction; only residuals get clustered for novelty.

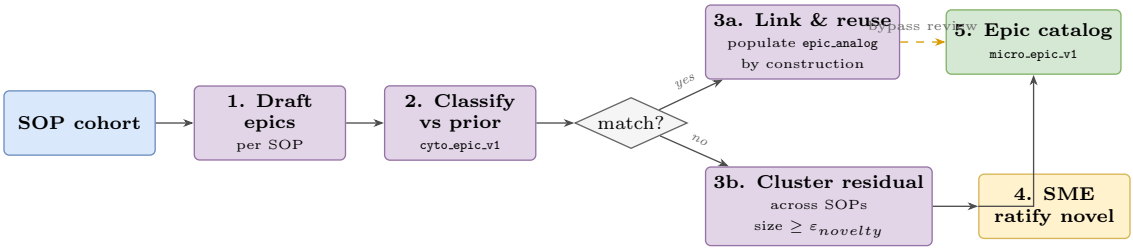


Figure 2: Conditioned Epic Extractor — the "match?" decision is algorithmic (not SME-gated; hence neutral coloring). Only the novel branch goes through SME ratification.

Result for the Micro POC: most Micro epics inherit Cyto's structure with explicit `epic_analog` populated by construction; only genuinely novel epics — e.g. Antibiotic Susceptibility Testing (AST), if it has no Cyto analog — emerge with cluster evidence. The dev team reads `micro_epic_v1` and immediately sees what's reused vs. new without having to do the mapping themselves.

A **shape** is the categorization the Story Extractor assigns to each Story. It tells the Validator which sub-rubric to apply, and tells downstream consumers (devs, the YAML emitter) how to interpret the AC. Per-SOP extraction picks one of four shapes per element. Cross-SOP synthesis only ever emits the `capability` shape.

`shape`	**Use when...**	**Validator rubric**
`capability`	SOP describes a configurable system feature you can imagine multiple labs/cultures parameterizing	MUST/SHALL on AC; testable; configurability boundaries explicit
`workflow-stage-split`	Same capability, but SOP describes different behavior at different workflow stages (before / during / after results)	Stage explicit in title; sibling stories enumerated and cross-linked
`configuration-instance`	Concrete data values for one culture / lab. *No new feature — just data.*	Typed values + verbatim citation. *No MUST/SHALL.*
`cleanup`	Modify or remove an existing artifact. *No new feature.*	Target named + before/after observable + regression risk

CAPABILITY
configurable feature
Title: *Specimen Rejection Rules Engine*
AC: *System SHALL allow rule definition with {predicate, threshold, action}.*
Rubric: *MUST/SHALL + testable.*

WORKFLOW-STAGE-SPLIT
one capability, by stage
Sibling titles:
PHI Update Before Work Has Started
PHI Update After Work Has Started
PHI Update After Results Reported
Rubric: *stage in title + cross-links.*

CONFIGURATION-INSTANCE
concrete data values
Title: *Urine centrifugation params*
Values: *{speed: 1500 g, duration: 5 min, temp: 4 °C}*
Rubric: *typed values + verbatim SOP citation.* No MUST/SHALL.

CLEANUP / CORRECTION
modify/remove existing artifact
Title: *Remove Result Bucket option from Add/Modify QT code screens*
Rubric: *target artifact named, before/after observable, regression risk acknowledged.*

Schemas — deep dive

Note on IDs: draft IDs (e.g. `epic_draft_3`, `story_draft_42`) are used in-pipeline; the Dependency Resolver replaces them with final Jira-shaped IDs (e.g. `EPIC-MICRO-001`, `STORY-MICRO-1042`) at Jira-push time.

Epic schema (v3.2)

```
{ id: "EPIC-MICRO-001", title: "Microbiology Specimen Receipt",
  description: "...", discipline: "microbiology",
  themes: ["G1","G5"], theme_catalog_version: "cyto_v1",
  epic_analog: { catalog_id: "cyto_epic_v1", epic_id: "EPIC-CYTO-007", equivalence: "identical" },
  inheritance_basis: { match_score: 0.78, shared_themes: ["G1"], auto_inherited: true } }
```

field	type	req'd	notes
id	string	yes	Jira-shaped Epic ID.
title	string	yes	Short capability-domain name.
description	string	yes	1–3 sentence scope statement.
discipline	enum	yes	microbiology | histology | cytology.
themes[]	array	yes	Multi-label tags; values drawn from the theme catalog named by theme_catalog_version.
theme_catalog_version	string	yes	*(v3.2)* The theme catalog these tags resolve in — e.g. cyto_v1 for Micro POC, histo_v1 for Histology.
epic_analog	object	no	*(v3.2 — replaces cyto_epic_analog)* Cross-catalog link: {catalog_id, epic_id, equivalence}. Populated by the conditioned Epic Extractor when match_score $\geq \tau_{match}$.
inheritance_basis	object	no	*(v3.2)* Provenance for inherited epics: {match_score, shared_themes[], auto_inherited}. Absent for novel epics; those carry novelty_basis instead.

Story schema

```
{ id: "STORY-MICRO-1042", epic_id: "EPIC-MICRO-001", shape: "capability",
  title: "Configurable specimen rejection rules engine", description: "...",
  acceptance_criteria: [{when, then, expected_value?}, ...],
  source_citations: [{doc_id, section, line_range}, ...],
  dependencies: ["STORY-MICRO-1018"],
  cross_links: ["STORY-MICRO-1043","STORY-MICRO-1044","STORY-MICRO-1045"],
  estimated_complexity: "M", edge_cases_handled: [...],
  status: "validated", source_chunks: ["chunk_001","chunk_017"] }
```

field	type	req'd	notes
id	string	yes	Jira-shaped Story ID.
epic_id	string	yes	FK to parent Epic.
shape	enum (4)	yes	Drives Validator routing.
title	string	yes	Concrete + unambiguous; for stage-split, must name the stage.
description	string	yes	User-story format for capability/stage-split; descriptive prose for config-instance/cleanup.
acceptance_criteria[]	array	yes	Capability/stage-split: MUST/SHALL phrasing. Config-instance: expected_value populated. Cleanup: before/after wording.
source_citations[]	array ⟨{doc_id, section, line_range}⟩	yes	Must resolve to indexed chunks; verbatim excerpt for config-instance.
dependencies[]	array ⟨Story ID⟩	no	Stories that must complete before this one is workable.
cross_links[]	array ⟨Story ID⟩	no	Capability ↔ concrete contributors; stage-split sibling group. Backbone of the Story DAG.
technical_hints	string	no	Dev-facing guidance.
estimated_complexity	enum S|M|L	yes	Heuristic, sprint sizing.
edge_cases_handled[]	array of strings	no	Surfaced corner cases.
status	enum	yes	draft | validated | ready.
source_chunks[]	array ⟨chunk ID⟩	yes	Audit trail to the retrieval substrate.

Cluster schema (intermediate, in Cross-SOP Synthesis)

Not persisted to Jira; lives only between gate #1 and gate #2 in memory.

```
{ cluster_id: "clust_rejection_3sop",
  member_story_ids: ["STORY-MICRO-1043","STORY-MICRO-1044","STORY-MICRO-1045"],
  source_sops: ["doc_micro_014_v3","doc_micro_022_v1","doc_micro_037_v2"],
  centroid_summary: "Reject specimen on a culture-specific predicate.",
```

```
  proposed_capability_title: "Configurable specimen rejection rules engine" }
```

A cluster qualifies for lifting **iff** `len(unique(source_sops)) >= 2`. Two stories from the same SOP do not qualify, even if they cluster tightly.

Validator output schema

```
{ story_id: "STORY-MICRO-1042", verdict: "revise",
  shape_declared: "capability", shape_verified: "capability",
  failed_checks: [
    {rule:"must_shall_present", severity:"error", message:"AC 1 lacks MUST/SHALL phrasing."},
    {rule:"ambiguous_quantifier", severity:"error", message:"AC 2 uses 'appropriate'."}
  ], attempt: 1 }
```

field	type	notes
`story_id`	string	The Story under review.
`verdict`	enum	`accept` (passes), `revise` (return to extractor with `failed_checks`, attempt < 2), or `escalate` (after 2 failed revisions; SME-routed).
`shape_declared`	enum	The shape the Story Extractor assigned.
`shape_verified`	enum	The shape the Validator believes is correct. If it differs from `shape_declared`, verdict is automatically `revise`.
`failed_checks[]`	array ⟨{rule, severity, message}⟩	Empty when `verdict = accept`.
`attempt`	integer	1 or 2 (after 2, verdict becomes `escalate`).

Theme catalog schema (v3.2)

```
catalog_id: histo_v1
parent_catalog: cyto_v1
discipline: histology
inherited_themes:
  - { id: G1, name: Pre-Analytic, source: cyto_v1, match_count: 142 }
  ...
novel_themes:
  - { id: H1, name: Tissue Processing, source: histo_v1,
    cluster_evidence: {n_chunks: 38, sample_excerpts: [...]}, sme_confirmed: true }
discarded_themes:
  - { id: G7, reason: "Folded into H1 (partial)", chunks_re_tagged: 12 }
unclassified_bucket: { count: 22, pct_of_total: 3.7 }
```

field	type	req'd	notes
`catalog_id`	string	yes	`<discipline>_v<N>`. Stories reference this in `theme_catalog_version`.
`parent_catalog`	string | null	yes	The prior the discovery was conditioned on. `null` only for the very first ever (`cyto_v1`).
`inherited_themes[]`	array	yes	Themes that survived from `parent_catalog` after Pass 1. Each has `source: <parent>` and a `match_count`.
`novel_themes[]`	array	yes	SME-ratified novelties from Pass 2. Each carries `cluster_evidence`. May be empty.
`discarded_themes[]`	array	opt	Parent themes the SME dropped. `chunks_re_tagged` records the Pass 4 fixup count.
`unclassified_bucket`	object	yes	G0 stats: {`count, pct_of_total, sample_excerpts, next_review_due`}. Always present. 5% alarm threshold.

Epic catalog schema (v3.2)

```
catalog_id: micro_epic_v1
parent_catalog: cyto_epic_v1
discipline: microbiology
inherited_epics:
```

```
  - { id: EPIC-MICRO-001, title: Specimen Receiving,
    epic_analog: {catalog_id: cyto_epic_v1, epic_id: EPIC-CYTO-014, equivalence: identical},
    inheritance_basis: {match_score: 0.78, shared_themes: [G1], auto_inherited: true} }
novel_epics:
  - { id: EPIC-MICRO-007, title: AST,
    epic_analog: null,
    novelty_basis: {n_draft_epics: 4, sme_confirmed: true} }
unclassified_drafts: { count: 3, pct_of_total: 4.1 }
```

The epic catalog versions independently from the theme catalog and uses the suffix `_epic_` to make the naming distinction explicit. The `epic_analog` field, when populated, carries a struct (not a bare string) so the catalog the analog points into is unambiguous.

ANALOGY map schema (v3.2)

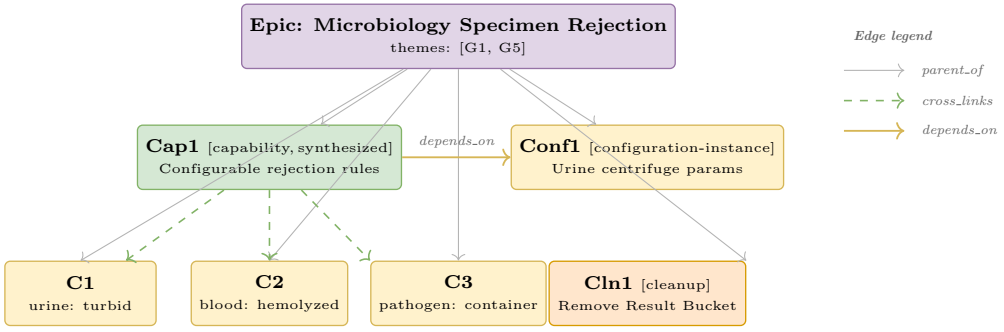
```
source_catalog: cyto_v1
target_catalog: histo_v1

theme_links:
  - { source: G1, target: G1, equivalence: identical }
  - { source: G7, target: H1, equivalence: partial,
    note: "G7 has no top-level peer in histo_v1 (discarded); concept partially absorbed by H1 for slide-prep instruments." }
  - { source: null, target: H2, equivalence: novel_in_target }
  ...(H3, H4)

epic_links:
  - { source: EPIC-CYTO-014, target: EPIC-HISTO-002, equivalence: identical }
```

`equivalence`	**meaning**
`identical`	Full equivalence; `match_score` above τ_{match}, SME-confirmed by spot-check.
`partial`	Loose mapping or scope difference; SME-confirmed each individually. Carries a `note`.
`discarded_in_target`	Source has no top-level peer in target.
`novel_in_target`	Target has no source ancestor (encoded as `source: null`).

Convention. Each source theme appears at most once as `source`. "Discarded with partial mapping" is a single `partial` entry whose note explains the discard — *not* two contradictory entries (`partial` plus `discarded_in_target`). Similarly, a target theme covered by a `partial` link does not also need a separate `novel_in_target` entry; the partial link already records its origin.

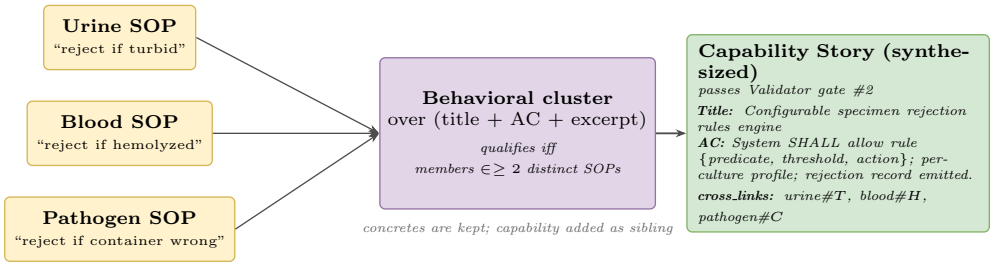
Story relationships (graph view)

The Story schema's `dependencies[]` and `cross_links[]` arrays serialize a directed acyclic graph. Walking it answers "what's blocked by X?" or "if I deprecate the synthesized capability story, which concretes need merging?"



edge	direction	meaning
`parent_of`	Epic → Story	Implicit Jira hierarchy (encoded in `epic_id`).
`cross_links`	Story ↔ Story	Bidirectional association. Capability ↔ concrete (synthesis); stage-split sibling sets.
`depends_on`	Story → Story	Directed prerequisite. The Dependency Resolver topologically orders the backlog for sprint planning.

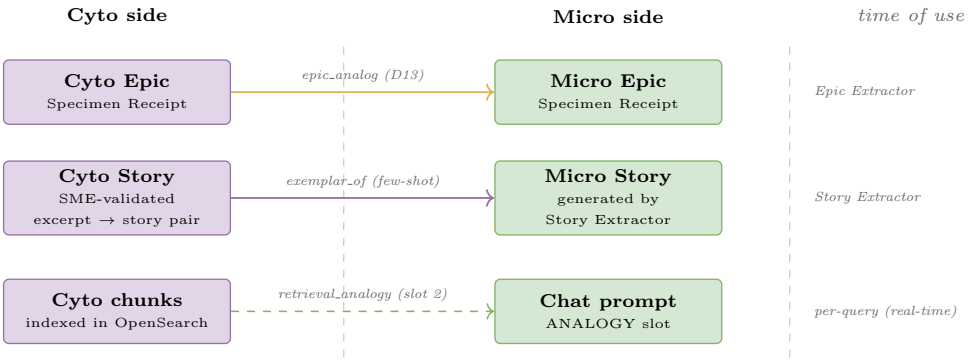
Cross-SOP synthesis — recurrence becomes capability



The recurrence threshold (≥ 2 distinct SOPs) is tunable. With three SOPs in the POC, this means anything seen in two of three lifts. The synthesized capability story passes Validator gate #2 before persistence.

Cyto → Micro lineage (graph view)

Three distinct mechanisms transfer information from Cyto to Micro. They operate at different times in the system's life cycle.



mechanism	direction	when it fires / what flows
`epic_analog`	Cyto Epic → Micro Epic	At Epic-Extractor time. Just an ID reference — recorded for SME traceability (D13).
`exemplar_of`	Cyto Story → Micro Story	At Story-Extractor time. The Cyto (excerpt → Story) pair lands in the prompt as an in-context example.
`retrieval_analogy`	Cyto chunks → chat prompt	Per-query, real-time. Retrieved Cyto excerpts populate the ANALOGY context section. *Adapt, don't copy.*

The same Cyto material thus serves three roles: a curated few-shot teaching set, an indexed corpus for live retrieval, and a structural reference at the Epic level.

v3.2 ANALOGY map (graph view) — explicit cross-discipline links

When a discipline's theme catalog diverges from its parent (Histology drops G7, adds H1–H4), the ANALOGY map records the cross-catalog correspondences explicitly. Each link is typed — the legend shows what each line style means.

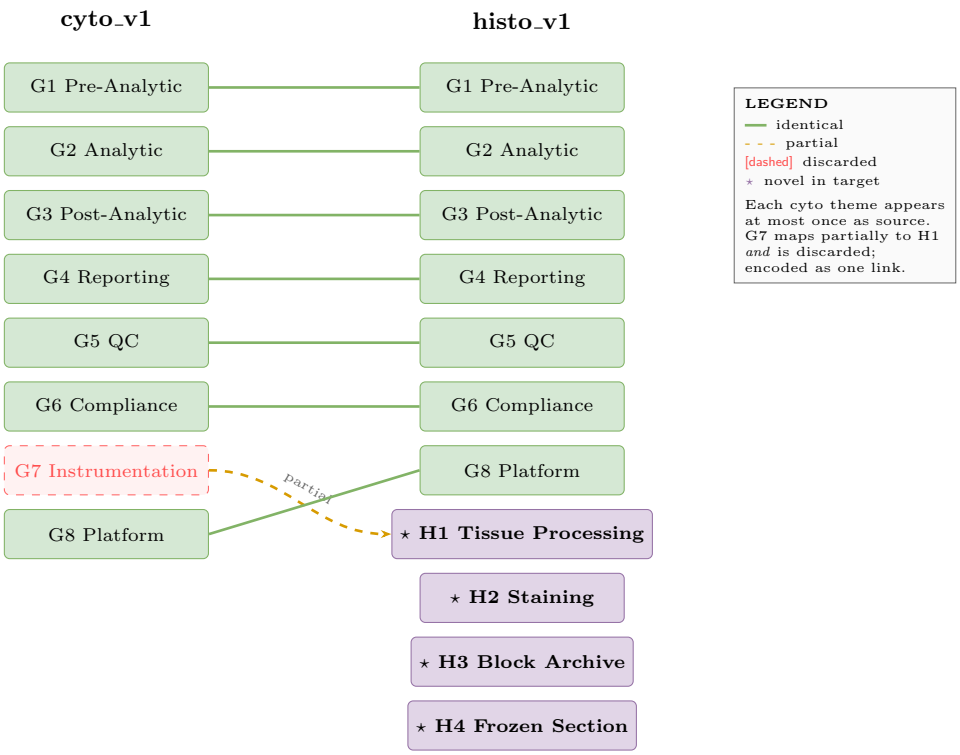
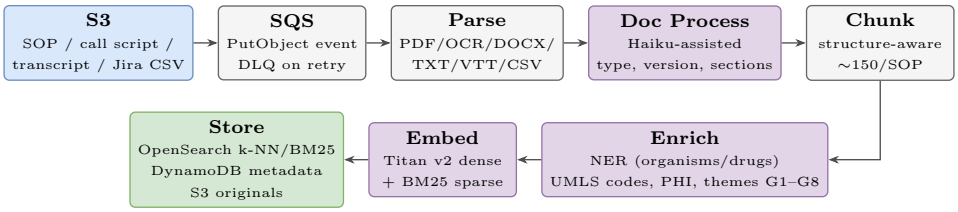


Figure 3: ANALOGY map (cyto_v1 ↔ histo_v1). 6 identical theme links, 1 partial (G7→H1), 4 novel-in-target (H1–H4). H1 has no separate `novel_in_target` edge because the partial G7→H1 link already records its origin.

The ANALOGY map drives the ANALOGY retrieval slot for cross-discipline queries: when a Histo question wants Cyto context, the map filters Cyto chunks down to themes that have a target-side correspondent.

Ingestion substrate — S3 to indexed chunks



I/O trace — one document through the pipeline (`SOP-MICRO-014_Gram-Stain_v3.docx`):

```
[parse out]
text: 28,420 chars / 18 pages
format: docx, ocr_used: false, parse_ms: 320
```

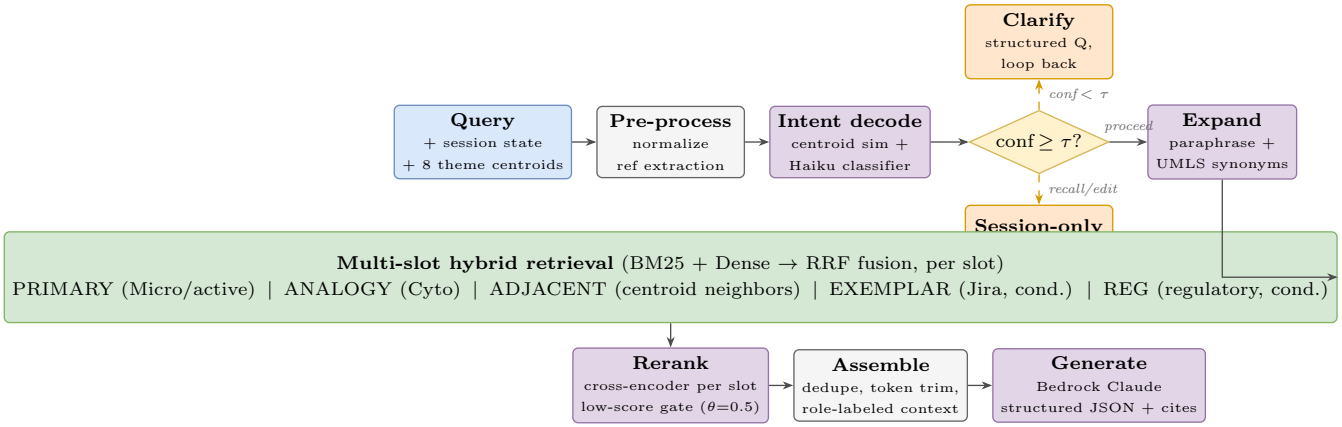
```
[doc process out]
doc_id: doc_micro_014_v3_abc12
discipline: Microbiology, version: v3
section_map: {Purpose:[200,450], Procedure:[721,2400],
AC:[2401,3100], QC:[3101,3650]}
```

```
[enrich out, chunk_001]
entities: {organisms:["Gram-pos cocci"],
specimens:["sputum"], codes:["87209"]}
phi_detected: false
themes: {G2: 0.84, G5: 0.41}
```

```
[embed out, chunk_001]
dense_vector: [0.0234, -0.1849, ..., 0.2891] (1024-d, Bedrock Titan v2)
bm25_tokens_enriched: ["slide","preparation","87209","smear","primary","source",...]
```

```
[store, sinks written]
PUT chunk_001 → OpenSearch k-NN index (vector)
PUT chunk_001 → OpenSearch BM25 index (sparse)
PUT chunk_001 → DynamoDB chunk_metadata (entities, themes, phi_status, version)
```

Retrieval substrate — per query, real-time



I/O trace — one query: "Generate epics for specimen rejection in microbiology"

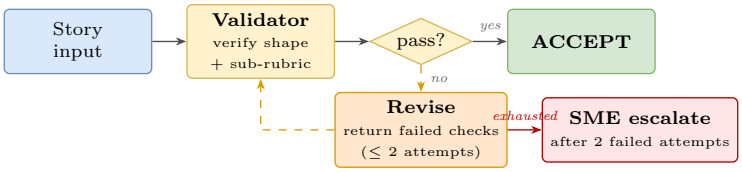
```
[intent decode out]
centroid_dist: {G1: 0.81, G5: 0.18}
classifier: {action: generate, themes: [G1, G5], discipline: Micro, granularity: epic,
refs: [], conf: 0.91}
decision: PROCEED (τ=0.7, 0.91 ≥ τ)
```

```
[retrieve out] 48 candidate chunks
  slot_1 PRIMARY (Micro/G1+G5): 20
  slot_2 ANALOGY (Cyto/G1+G5): 10
  slot_3 ADJACENT (Micro/G2 neighbors): 10
  slot_4 EXEMPLAR (Jira, generate/\epic): 5
  slot_5 REG (CLSI on G5): 3
```

```
[rerank + assemble out]
trim: 8/4/3/3/2 = 20 chunks (all slots ≥ θ=0.5)
prompt sections: PRIMARY, ANALOGY, ADJACENT, EXEMPLAR, REG
+ session drafts + output schema
total tokens: 6,840
```

```
[generate out] structured JSON
{ id: "epic_draft_3", # draft ID, replaced by
              # Dep. Resolver before Jira push
  title: "Specimen Rejection Workflow",
  themes: ["G1","G5"],
  citations: ["s3://.../SOP-MICRO-007 §AC",
             "s3://.../SOP-CYTO-021 §Procedure",
             "CLSI M22 §3.4"],
  stories: [{title: "Validate transport temp", ...}, ...] }
```

Validator — shape-aware revise/escalate flow



Sub-rubrics by shape (cheat sheet):

- **Capability:** MUST/SHALL on every AC; testable; no ambiguous quantifiers ("appropriate", "relevant", "necessary"); configurability boundaries called out; SOP citation present and resolves; complexity S/M/L; edge cases enumerated.
- **Stage-split:** stage explicit in title; entry/exit observable per stage; siblings enumerated and cross-linked; transition contract clear; boundary cases handled.
- **Configuration-instance:** typed concrete values (units / enums / nullable rules); target table or profile named (e.g. `cultures/urine.yaml#centrifugation`); SOP excerpt cited verbatim; scope (culture/lab) stated. *MUST/SHALL not required.*
- **Cleanup:** target artifact named (id, screen path, file path); before/after observable; regression risk acknowledged; DoD includes a smoke test or QA path. *MUST/SHALL not required.*

What ships — Output A (Jira) and Output B (YAML)

Output A — Jira artifacts. The connected Epic / Story DAG (graph above), pushed to Jira via REST. The graph is what the dev team works against; the YAML profile (Output B) is what their code consumes at runtime.

Output B — Per-culture YAML configuration profile. One YAML file per culture in scope. Carries the concrete typed values that `configuration-instance` Stories' AC referenced. Profiles are versioned, dated, and cite back to the SOPs they draw from. **The Story body never duplicates these values** — it references them by `related_story_ac`, so a single source of truth is preserved.

The profile is partitioned by workflow stage so an engineer can find the right block at a glance:

- `pre_analytic` — before the specimen reaches the bench: collection, transport, rejection predicates.

- `analytic` — the bench work itself: centrifugation, inoculation, incubation.
- `post_analytic` — after the result: reporting thresholds, downstream notifications.

Full profile structure (`cultures/urine.yaml`)

```yaml
profile:
  culture: urine          # urine | blood | target_pathogen
  version: 3
  effective_date: 2026-04-15
  source_sops: [doc_micro_014_v3, doc_micro_022_v1]

pre_analytic:
  collection:
    container: {value: "sterile cup", units: enum, citation: "doc_micro_014_v3 §1.2 L8"}
    volume_min: {value: 10, units: mL, citation: "doc_micro_014_v3 §1.2 L11"}
  transport:
    temp_range: {min: 2, max: 8, units: degC, citation: "doc_micro_014_v3 §1.4 L18-19"}
    time_max: {value: 4, units: h, citation: "doc_micro_014_v3 §1.4 L20"}
  rejection_predicates:
    - id: REJ_TURBID
      condition: "turbidity > threshold"
      threshold: {value: 200, units: NTU}
      action: reject
      citation: "doc_micro_014_v3 §1.5 L24"

analytic:
  centrifugation:
    speed: {value: 1500, units: g, citation: "doc_micro_014_v3 §3.2 L41-43"}
    duration: {value: 5, units: min, citation: "doc_micro_014_v3 §3.2 L44"}
    temp: {value: 4, units: degC, citation: "doc_micro_014_v3 §3.2 L45"}
  inoculation:
    media:
      - {type: BAP, volume_uL: 10, citation: "doc_micro_014_v3 §4.1 L52"}
      - {type: MAC, volume_uL: 10, citation: "doc_micro_014_v3 §4.1 L53"}
    incubation: {temp: 35, atmosphere: "5% CO2", duration_h: 18,
        citation: "doc_micro_014_v3 §4.3 L60"}

post_analytic:
  reporting:
    cfu_threshold: {value: 100000, units: CFU_per_mL,
        citation: "doc_micro_014_v3 §6.1 L88"}

related_story_acs: # back-references to Stories that consume this profile
  - STORY-MICRO-1042#AC-2 # Configurable rejection rules engine
  - STORY-MICRO-1077#AC-1 # Centrifugation step
```

How a Story's AC references the profile

A `configuration-instance` Story's AC carries the citation; the profile is what gets read by code. Example — `STORY-MICRO-1077`:

```yaml
acceptance_criteria:
  - when: "urine profile loaded"
    then: "centrifuge step uses cultures/urine.yaml#analytic.centrifugation"
    expected_value: "speed=1500g, duration=5min, temp=4degC"
```

Where to put a value: Story body vs. profile

If the value is...	Put it in...
The same across cultures and parameterizable in the rules engine	Story body as parameter spec (capability shape)
Different per culture / per lab	YAML profile (configuration-instance shape)
A literal constant in the rejection logic	Profile, with the Story's AC pointing at it

Cross-cutting

KMS encryption at rest and in transit; IAM least-privilege with VPC endpoints (Bedrock calls stay in-VPC); DLQ for parse / OCR / embed failures; CloudWatch metrics and alarms (e.g. OCR fail-rate > 5%); CloudTrail audit log on every API call; idempotent ingestion via content-hash dedupe.

Generalization (v3.2): when Histology arrives, Theme Discovery runs once as pre-flight to warm-start `histo_v1` from `cyto_v1` (most themes inherited; novel ones SME-ratified). The main pipeline then runs unchanged in shape, with the Epic Extractor in conditioned mode bootstrapping `histo_epic_v1` from `cyto_epic_v1`. No model retraining at any stage.